ZIJIAN GUO (Steven)

D.O.B.: May. 28, 2007 TEL: 86-13853128177 Email: guo05@berkeley.edu

Education

- **Jinan Xinhang Experimental Foreign Language School September 2022 — May 2025**

Standard Examinations: TOEFL (110), GCE A Level Predicted Scores: Physics A*, Further Mathematics A*, Economics A*, Math A*

IGCES Scores Mathematics A* Physics A* Economics A Chemistry A English As a Second Language B

- **Brown University Precollege July 2024–August 2024**

Took courses in May The Force Be With You: Physics for the Ages

- **High school affiliated to Shandong University September 2019 — June 2022**

Honors & Awards

- United States Academic Decathlon（in campus） The third prize 2022
- Australian Mathematics Competition The third prize 2022
- Silver (Global Ranking), British Physics Olympiad 2023-24 2023
- Gold Certificate, UKMT Senior Mathematical Challenge 2023
- Super Econ! Regional Overall Team Award in Pre, National Economics Challenge China Finals 2023
- Regional Overall Team Bronze Award in Pre, National Economics Challenge China Preliminary 2023
- Distinction Top 5%, 130.5/150, American Mathematics Competition 12B 2023
- Gold Certificate, UKMT Intermediate Mathematical Challenge 2023
- Galois Contest Distinction 2023
- The Galois Mathematics Competition distinction 2023

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- Math Kangaroo Competition The bronze award 2023
- Top 16%, Warton Global High School Investment Program Exploration 2024
- Meritorious, 2023 High School Mathematical Contest In Modeling 2024
- National Silver, AAPT Physics Bowl 2024

Research Program

- **Research paper (The Structural Optimization and Space Layout Study of Steel-Steel Composite Transfer Trusses in High-Rise Buildings)**
  - **Detailed analysis of how the steel-steel composite transfer trusses distribute loads across the structure, considering both vertical and horizontal forces.**
  - **Comparison of the performance of these case studies with traditional truss designs, highlighting the improvements in efficiency and space utilization.**
- **Research Program + dissertation on physics and architecture**

Internship & Volunteer

- **High School Library assistant September 2022——present**

Provided front desk support: greeted students and faculty, answered inquiries, and guided visitors to resources.

Organized and shelved books by Dewey Decimal System and performed regular inventory checks.

Assisted in setting up displays and promoting reading events to encourage student engagement.

Maintained a quiet and welcoming environment conducive to study and reading.

Managed book check-ins and check-outs using the school's library system, maintaining accurate circulation records.

- **Tutor & Volunteer, Advanced Placement Level Program Chinese September 2022**

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——December 2022

Teach Chinese to 3 American students online; design courseware; arrange course activities and themes depend on hot topics such as "Chinese Cuisine" and "Modernization".

➢ Docent – Shandong Provincial Museum Program September 2022——present

While familiarizing myself with and memorizing the historical and cultural information of "prehistoric cultural relics," I focused on key artifacts such as Neolithic pottery, bone tools, and stone implements. For example, I delved into the jade dragon ornaments from the Hongshan culture and the painted pottery designs from the Yangshao culture. These relics not only reflect the craftsmanship of ancient humans but also reveal their social structures and religious beliefs.

To better serve different audiences, I practiced various expression techniques and explanatory approaches tailored to different age groups and interests. For instance, when guiding young children, I used engaging stories and interactive questions to explain how prehistoric people used simple tools for hunting and daily life, sparking their curiosity. For adult audiences, I provided more in-depth discussions on the cultural context and the archaeological discovery process behind the relics, such as explaining the craftsmanship and uses of certain stone tools to showcase the intelligence and creativity of ancient humans.

Additionally, I experimented with leading museum visits for various age groups, continuously adjusting and refining my explanations based on their feedback. For example, when guiding senior groups, I adopted a slower pace and included more background information to help them better understand the historical significance of the relics. Through these practical experiences, I gradually learned how to provide targeted cultural heritage explanations that cater to different audience needs, ensuring that each visitor gains valuable insights and develops a deep interest in prehistoric cultures.

➢ Initiator, Leader, Director – Physics Course Project September 2022——present

In launching the STEM subject popularization and physics teaching project, I aimed to create a platform that could make complex scientific concepts more accessible and engaging for a broader audience. The project involved organizing a team of members to record teaching videos that were then uploaded on platforms like YouTube and Red Book. These videos targeted a diverse audience, including students who might not have access to quality STEM education, as well as those looking to deepen their understanding of physics and related subjects.

The primary goal of uploading these videos was to reach students across different regions, providing them with valuable educational resources. The content was designed to be both informative and engaging, with the intention of sparking an interest in STEM subjects among viewers. By making these resources widely available online, we hoped to inspire curiosity and a love for learning in students who might not have had the opportunity to explore these subjects in depth.

Our decision to visit Liubu No. 1 Middle School for in-person teaching was driven by a desire to directly engage with students in a more hands-on environment. The purpose of this outreach was twofold: first, to introduce young students to the fascinating world of physics by identifying and nurturing their interests in specific areas of the subject; and second, to enhance our own passion for learning by sharing our knowledge and witnessing the impact it had on others.

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During the teaching sessions, we focused on finding out what aspects of physics excited the students most and tailored our lessons to those interests. This approach not only made the learning experience more enjoyable for the students but also helped us to better understand how to communicate complex ideas in a relatable way. The experience of teaching and engaging with these students was incredibly rewarding and further deepened our own enthusiasm for STEM subjects, reinforcing our commitment to making science education more accessible and engaging for everyone.

Extracurricular Activities

➢ Founder, Leader – Green Design Society September 2022——Present

Established a community with the original intention of "making environmentally friendly and practical products with green and renewable materials". The motto is "use less ues more". It's about using as few items as possible to create as much space as possible to use the direction. Invite the school physics teacher to be the instructor. Design new tools based on waste materials according to some needs in the campus, such as: sofa made of mineral water bottles, eco-friendly cat house.

➢ Founder, Leader – Physics Top-Lab Program September 2023——present

Inspired by the Waste Physics Lab, an organization was established to shoot physics experiment videos, record relevant experiments based on the physics knowledge students are learning, using a humorous video style to introduce obscure physics concepts, so that students can easily watch them during their lunch break. Upload the video to the website (YouTube, Bilibili, Tiktok) at a later stage.

➢ Designer – Jinan Museum “Cultural and Creative Products

As a designer for the Jinan Museum’s “Cultural and Creative Products” project, I volunteered to create a cultural design inspired by the museum’s collection. Specifically, I developed a cartoon-style image based on the museum's treasured Western Han Painted Negative Pot Pottery Bird. This design aims to both reflect the historical and artistic significance of the artifact and appeal to a modern audience with its engaging and friendly style.

1. Inspiration Collection: Before starting the design, I thoroughly researched the Western Han Painted Negative Pot Pottery Bird, focusing on its unique painted and carved details. I used reference materials from the museum and

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various documents to ensure accuracy and authenticity in my design.

2. Sketching: I began by sketching several concepts, exploring how to translate the traditional pottery bird into a contemporary cartoon style. These sketches emphasized capturing the basic shape and distinctive features of the bird, such as its rounded body, decorative feather patterns, and exaggerated eyes and expressions, to add charm and visual appeal.

3. Digital Design: Adopting a clean and expressive cartoon style, I used graphic design software to create a digital version of the image. I employed bright colors and smooth lines to highlight the pottery bird's distinctive elements, such as its unique painted details and carving textures. I ensured that these features were clearly visible even in the cartoon style, maintaining the historical and cultural essence of the artifact.

4. Design Refinement and Optimization: After completing the initial design, I made several adjustments and refinements to ensure that the image performed well across different applications. This involved tweaking color saturation, line thickness, and overall composition to ensure the final cultural design was both aesthetically pleasing and true to the original artifact's spirit.

5. Final Presentation: Once the design was finalized, I submitted it to the Jinan Museum, where it received positive feedback. This cultural design was not only featured as one of the museum's creative products but was also used in related promotional materials, enhancing the museum's cultural outreach efforts.

Through this project, I deepened my understanding of the Western Han Painted Negative Pot Pottery Bird and used modern design techniques to present traditional culture in a new and engaging way, revitalizing ancient artifacts for contemporary audiences.

Skills & Hobbies

Badminton, Tennis, Guitar, Swimming, photographing, Hiking, collection model cars,